Lecture30_MA5104

12 October 2026 10:42

Lecture30_ MA5104

MA 5104: Statistical Inference and Learning

Module 4: Learning

Lecture 30: Regression, Regularization, and Model Selection

Dr. Rakhi Singh
Department of Mathematics
IIT Madras

Fall 2026

Statistical Properties of Least Squares

$\hat{\beta} \rightarrow$ OLS estimator

Assumptions: $\epsilon \sim \mathcal{N}(0, \sigma^2 I)$. Then, the distribution of estimator:

$$\hat{\beta} \sim \mathcal{N}\left(\beta, (X^T X)^{-1}\sigma^2\right)$$

$\hat{\beta}$ is BLUE = (Best Linear Unbiased Estimator)

Variance estimate:

$$\hat{\sigma}^2 = \frac{1}{n-p-1}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$V(\hat{\beta})$ is minimum among the class of all unbiased estimators

Properties: $\mathbb{E}[\hat{\beta}] = \beta$ (unbiased), $\mathbb{E}[\hat{\sigma}^2] = \sigma^2$.

Additional results: $(n-p-1)\hat{\sigma}^2 \sim \sigma^2 \chi^2_{n-p-1}$

Use confidence intervals for β_j and perform hypothesis testing.

Optimization via Gradient Descent

Gradient Derivation (Step-by-Step)

Loss: $L(\beta) = \frac{1}{n}(y - X\beta)^T(y - X\beta)$

Expand:

$$= \frac{1}{n}(y^T y - 2\beta^T X^T y + \beta^T X^T X\beta)$$

Differentiate:

$$\nabla L(\beta) = -\frac{2}{n}X^T y + \frac{2}{n}X^T X\beta$$

$$= -\frac{2}{n}X^T(y - X\beta)$$

Update rule:

$$\beta_{t+1} = \beta_t - \eta\nabla L(\beta_t)$$

Substitute gradient:

$$\beta_{t+1} = \beta_t + \frac{2\eta}{n}X^T(y - X\beta_t)$$

Optimization via Gradient Descent

Gradient Descent Update and Interpretation

Rewrite:

$$\beta_{t+1} = \beta_t + \frac{2\eta}{n}X^T r_t, \quad r_t = y - X\beta_t$$

Residual

Interpretation:

- Update driven by correlation between features and residuals
- If $X_j^T r_t \neq 0$, coefficient β_j is adjusted

p

Connection to normal equations:

$$X^T(y - X\beta^*) = 0$$

- Gradient descent seeks β such that residual is orthogonal to column space

Convergence:

- Depends on eigenvalues of $X^T X$
- Poor conditioning $\Rightarrow$ slow convergence (zig-zag path)

Model Selection

The Model Selection Problem

With p predictors there are 2^p possible subsets.

p	subsets	at 1 ms each
10	1,024	1 second
20	10^6	17 minutes
40	1.1×10^{12}	35 years

We do not yet know how to decide which predictors go in. Two obstacles:

1. **Too many candidates to try.** We cannot fit 2^p models, so we need a way of looking at a small number of promising ones.
2. **Scoring is slow.** Cross-validating even those few means refitting each one many times. Is there a quicker way to compare models?

There is a different route: keep all the predictors and shrink their coefficients towards zero. No search, no comparison of subsets — just one tuning parameter.

Model Selection

Subset Selection Framework

Goal:

$$\min_{\beta} \|y - X_S\beta_S\|^2$$

All subsets $p_{C_1}, p_{C_2}, \ldots p_{C_p} = 2^{p-1}$

Forward selection

y — x_1, x_2, x_{10} } step1

$y - x_2$ — x_1, x_3, x_7, x_{10} } step2

$y - x_2 + x_7$ } step3

$y = x_2 + x_7$ is my final model

Best model could be $y = \beta x_1 + \gamma x_5 + \epsilon$

$y \sim x_2 + x_5$ cannot be the best model

Goal:

$$\min_{\beta} \|y - X_S\beta_S\|^2$$

where $S \subset \{1, \ldots, p\}$ is a subset of variables.
Key idea:

- Fit least squares on selected variables only
- Eliminate others (set coefficients to zero)

Produces:

- Sequence of models indexed by subset size $k = |S|$
- Increasing $k \to$ decreasing training error

Key problem:

How to choose optimal subset size k?

Best Subset Selection

For each $k = 0, 1, \ldots, p$:

$$\text{Find } S_k = \arg\min_{|S|=k} \|y - X_S\beta_S\|^2$$

Properties:

- Searches all 2^p subsets
- Returns best model for each size k

Important:

- Training RSS decreases with k
- Cannot choose k using training error alone

Tradeoff:

- Large $k \to$ low bias, high variance
- Small $k \to$ high bias, low variance

Limitation:

Computationally infeasible for large p

Stepwise Methods

Forward Stepwise

- Start with intercept
- Add variable that most reduces RSS
- Produces nested sequence

Backward Stepwise

- Start with full model
- Remove least useful variable
- Requires $n > p$

Forward Stagewise

- Adds very small increments to coefficients
- Extremely conservative updates
- Useful in high-dimensional settings

From Subset Selection to Regularization

Subset selection can reduce bias, but often has high variance.

Issue: Model selection over 2^p subsets $\Rightarrow$ unstable

Idea: **Shrinkage (continuous alternative)**

$$\min_{\beta} \|y - X\beta\|^2 + \lambda\Omega(\beta)$$

$\lambda \sum \beta_j^2$ or $\lambda \sum |\beta_j|$

Key effects:

- Shrinks coefficients toward zero
- Reduces variance without hard selection
- Produces a continuum of models (controlled by λ)

Goal:

Improve prediction by bias–variance tradeoff

Ridge vs LASSO: Key Difference

Ridge (L2 penalty)

$$\Omega(\beta) = \sum \beta_j^2$$

- Shrinks coefficients continuously
- No exact zeros

LASSO (L1 penalty)

$$\Omega(\beta) = \sum |\beta_j|$$

- Produces sparse solutions
- Performs variable selection

Key contrast: Ridge: shrinkage LASSO: shrinkage + selection

Ridge vs lasso in picture

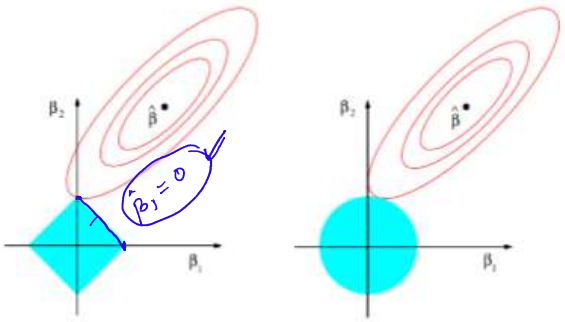


FIGURE 3.11. *Estimation picture for the lasso (left) and ridge regression (right). Shown are contours of the error and constraint functions. The solid blue areas are the constraint regions $|\beta_1| + |\beta_2| \le t$ and $\beta_1^2 + \beta_2^2 \le t^2$, respectively, while the red ellipses are the contours of the least squares error function.*

Ridge Regression: Formulation

Objective:

$$\hat{\beta}_{ridge} = \arg\min_{\beta} \|y - X\beta\|^2 + \lambda \sum \beta_j^2$$

Equivalent constrained form:

$$\min_{\beta} \|y - X\beta\|^2 \quad \text{s.t.} \quad \sum \beta_j^2 \leq t$$

Closed-form solution:

$$\hat{\beta}_{ridge} = (X^T X + \lambda I)^{-1} X^T y$$

Biased

Effect:

- Adds $\lambda I \rightarrow$ stabilizes inversion
- Handles multicollinearity

Profile plot: Ridge

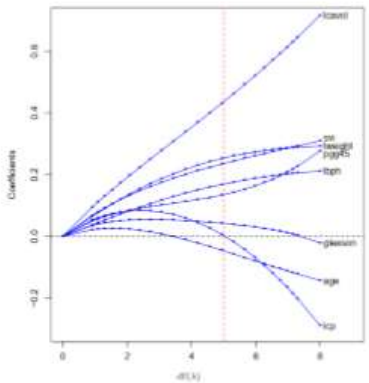


FIGURE 3.8. *Profiles of ridge coefficients for the prostate cancer example, as the tuning parameter λ is varied. Coefficients are plotted versus $df(\lambda)$, the effective degrees of freedom. A vertical line is drawn at $df = 5.0$, the value chosen by cross-validation.*

LASSO: Formulation and Sparsity

Objective:

$$\hat{\beta}_{lasso} = \arg\min_{\beta} \|y - X\beta\|^2 + \lambda \sum |\beta_j|$$

Equivalent constraint:

$$\sum |\beta_j| \leq t$$

Key property:

- Non-differentiable at $\beta_j = 0$
- Leads to exact zeros in coefficients

Result:

Sparse model $\Rightarrow$ automatic variable selection

Lasso: Profile plots

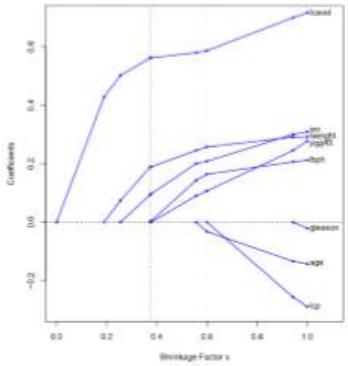


FIGURE 3.10. *Profiles of lasso coefficients, as the tuning parameter t is varied. Coefficients are plotted versus $s = t/\sum_1^p |\beta_j|$. A vertical line is drawn at $s = 0.36$, the value chosen by cross-validation. Compare Figure 3.8 on page 65; the lasso profiles hit zero, while those for ridge do not. The profiles are piece-wise linear, and so are computed only at the points displayed; see Section 3.4.4 for details.*

LASSO: Geometry and Behavior

Constraint region:

- L2 (ridge): sphere
- L1 (lasso): diamond

Key insight:

- Corners of L1 constraint $\rightarrow$ coefficients exactly zero

Comparison:

- Ridge: distributes weight across correlated variables
- LASSO: selects one, discards others

Tradeoff:

- Ridge $\rightarrow$ lower variance
- LASSO $\rightarrow$ interpretable sparse model

Estimates

TABLE 3.4. *Estimators of $\hat{\beta}_j$ in the case of orthonormal columns of* $\mathbf{X}$. *M and λ are constants chosen by the corresponding techniques; sign denotes the sign of its argument (± 1), and x_+ denotes "positive part" of x. Below the table, estimators are shown by broken red lines. The 45° line in gray shows the unrestricted estimate for reference.*

Estimator	Formula
Best subset (size M)	$\hat{\beta}_j \cdot I(\lvert\hat{\beta}_j\rvert \geq \lvert\hat{\beta}_{(M)}\rvert)$
Ridge	$\hat{\beta}_j/(1+\lambda)$
Lasso	$\mathrm{sign}(\hat{\beta}_j)(\lvert\hat{\beta}_j\rvert - \lambda)_+$

